# FINAL PROJECT REPORT

## Project Title: AI-Based 5G KPI Anomaly Detection Dashboard

## Introduction

Telecommunication networks continuously generate large volumes of Key Performance Indicators (KPI) data to monitor network health and quality of service. Manual analysis of these KPIs is time-consuming and may delay the identification of network issues. Artificial Intelligence (AI) and Machine Learning (ML) provide an efficient way to automate anomaly detection, enabling proactive maintenance and improved network reliability.

This project presents an AI-based solution for detecting anomalies in 5G telecom KPI data using the Isolation Forest algorithm. The system processes telecom KPI datasets, performs feature engineering, detects abnormal network h=behavior, and visualizes the results through an interactive Streamlit dashboard.

## Problem Statement

Telecom operators manage thousands of cell towers that continuously generate KPI data such as signal strength, throughput, latency, and packet loss. Monitoring this data manually is inefficient and increases the risk of delayed fault detection.

The objective of this project is to develop an intelligent anomaly detection system that automatically identifies unusual KPI patterns and assists network engineers in monitoring the health of the telecom network.

## Project Objectives

The primary objectives of this project are:

- Perform data preprocessing on telecom KPI data.
- Generate meaningful engineered features.
- Detect anomalies using the Isolation Forest algorithm.
- Tune model hyperparameters for improved performance.
- Build an interactive Streamlit dashboard.
- Visualize KPI trends and anomaly distributions.
- Provide a scalable framework for telecom network monitoring.

## Dataset Description

The project uses a telecom KPI dataset containing network performance metrics collected from multiple 5G cells.

### Dataset Features

| Feature | Description |
|---|---|
| Cell_ID | Unique identifier of the telecom cell |
| Timestamp | Date and time of KPI measurement |
| RSRP | Reference Signal Received Power |
| SINR | Signal-to-Interference-plus-Noise Ratio |
| Latency | Network delay |
| Throughput | Data transmission speed |
| Packet_Loss | Percentage of lost packets |
| Connected_Users | Number of active users |
| Signal_Quality_Score | Engineered feature representing signal quality |
| Network_Load | Engineered feature representing network utilization |
| Network_Health_Index | Engineered feature representing overall network condition |

## Data Preprocessing

The preprocessing stage prepared the raw dataset for machine learning.

The following operations were performed:

- Loaded for telecom KPI dataset
- Checked for missing values
- Removed duplicate records
- Converted the Timestamp column to datetime format
- Validated dataset consistency
- Saved the cleaned dataset for further processing

This step unsured that the dataset was complete, consistent, and suitable for machine learning.

### Feature Engineering

Three additional features were created to improve anomaly detection.

### Signal Quality Score

Combined signal strength (RSRP) and signal quality (SINR) to represent overall radio performance.

### Network Load

Represents the utilization of the network using the number of connected users.

### Network Health Index

Combined throughput, latency, and packet loss into a signal metric representing the overall health of the network.

These engineered features provide additional information to the anomaly detection model beyond the original KPIs.

## Machine Learning Model

The project uses the Isolation Forest algorithm for anomaly detection.

### Reason for Selecting Isolation Forest

- Unsupervised learning algorithm
- Does not require labeled training data
- Efficient for high-dimensional datasets
- Well suited for anomaly detection
- Performs effectively on telecom KPI data

### Model parameters

| Parameter | Value |
|---|---|
| Algorithm | Isolation Forest |
| n_estimators | 100 |
| contamination | 0.05 |
| random_state | 42 |

## Hyperparameter Tuning

Different combinations of the following parameters were evaluated:

- Number of Trees (n_estimators)
- Contamination Rate

The final configuration selected was:

- n_estimators=100
- contamination=0.05
- random_state=42

This configuration provided stable anomaly detection while maintaining reasonable computational efficiency.

## Model Evaluation

The trained model was evaluated using the processed telecom KPI dataset.

The evaluation included:

- Total records analyzed
- Number of normal records
- Number of anomalies detected
- Percentage of anomalies
- Anomaly distribution visualization

If reference labels are available, additional metrics such as Accuracy, Precision, Recall, and F1 Score can also be calculated. In this project, the Isolation Forest model was primarily evaluated as an unsupervised anomaly detector.

### Streamlit Dashboard

An interactive dashboard was developed using Streamlit.

The dashboard provides:

- Telecom KPI summary statistics
- Cell ID filtering
- Throughput trend visualization
- SINR distribution
- RSRP versus Throughput scatter plot
- Anomaly distribution pie chart
- Table displaying detected anomalies
- CSV download functionality

The dashboard enables users to monitor network health in an intuitive and interactive manner.

## Project Workflow

Raw telecom KPI dataset

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Data preprocessing

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Feature engineering

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Feature scaling

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Isolation forest model

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Anomaly detection

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Model evaluation

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Streamlit dashboard

## Results

The project successfully:

- Cleaned and processed telecom KPI data
- Generated engineered features to improve analysis
- Trained an Isolation Forest anomaly detection model
- Detected abnormal KPI behavior
- Performed hyperparameter tuning
- Built an interactive Streamlit dashboard
- Produced visualizations for telecom network monitoring

The solution demonstrates how machine learning can assist telecom engineers in identifying abnormal network conditions quickly and efficiently.

## Future Scope

The project can be extended with several advanced capabilities, including:

- Real time KPI streaming
- Deel learning based anomaly detection using LSTM or Autoencoders
- Cloud deployment using AWS, Azure, or Google Cloud
- Integration with telecom OSS/BSS systems

- Automated alert generation through email or messaging services
- Explainable AI (XAI) techniques for interpreting anomaly predictions
- Support for multi-user authentication and role-based access

## Conclusion

The AI-Based 5G KPI Anomaly Detection Dashboard demonstrates the practical application of machine learning for telecom network monitoring. By combining data preprocessing, feature engineering, Isolation Forest anomaly detection, and an interactive Streamlit dashboard, the project provides a structured approach to identifying abnormal network behavior.

The developed solution is scalable, easy to extend, and serves as a strong foundation for future AI-driven telecom network management systems.